

Blockchain-based digital product passports for EU ESPR compliance and consumer trust in textile and luxury retail.

Retail × Product traceability and digital product passport × Blockchain and DLT · OS040 v2 · refreshed 2026-08-18

ATTRACTIVENESS	RIGHT TO WIN	COMPETITION	SERVICEABLE MARKET	HORIZON	PORTFOLIO
70 is the world moving	20 can we play, can we win	LOW 0 named in the evidence	€981m modelled evidence · per year	NEXT regulation adopted or propose...	L3 19 signals

Attractiveness and right to win are scored and displayed separately and are never combined (SC-12). Competition is a fourth quantity, not a discount on either.

The opportunity

This opportunity is about deploying blockchain-based Digital Product Passports (DPPs) for textile and luxury retailers to comply with the EU's Ecodesign for Sustainable Products Regulation (ESPR) and to build consumer trust through verified product provenance. The EU has launched the DPP Registry and technical standards are now in place, making this a timely, regulatory-driven need for retailers selling into the EU.

Market opportunity

Method	Total (TAM)	Serviceable (SAM)	Obtainable (SOM)	Confidence
Bottom-up: enterprises x adoption x contract value	€1.8bn €344m – €8.4bn	€981m €185m – €4.5bn	€1.2m €232k – €5.7m	modelled

All figures are annual, in EUR. Ranges compound the uncertainty of each factor below; the base case is the middle column of each.

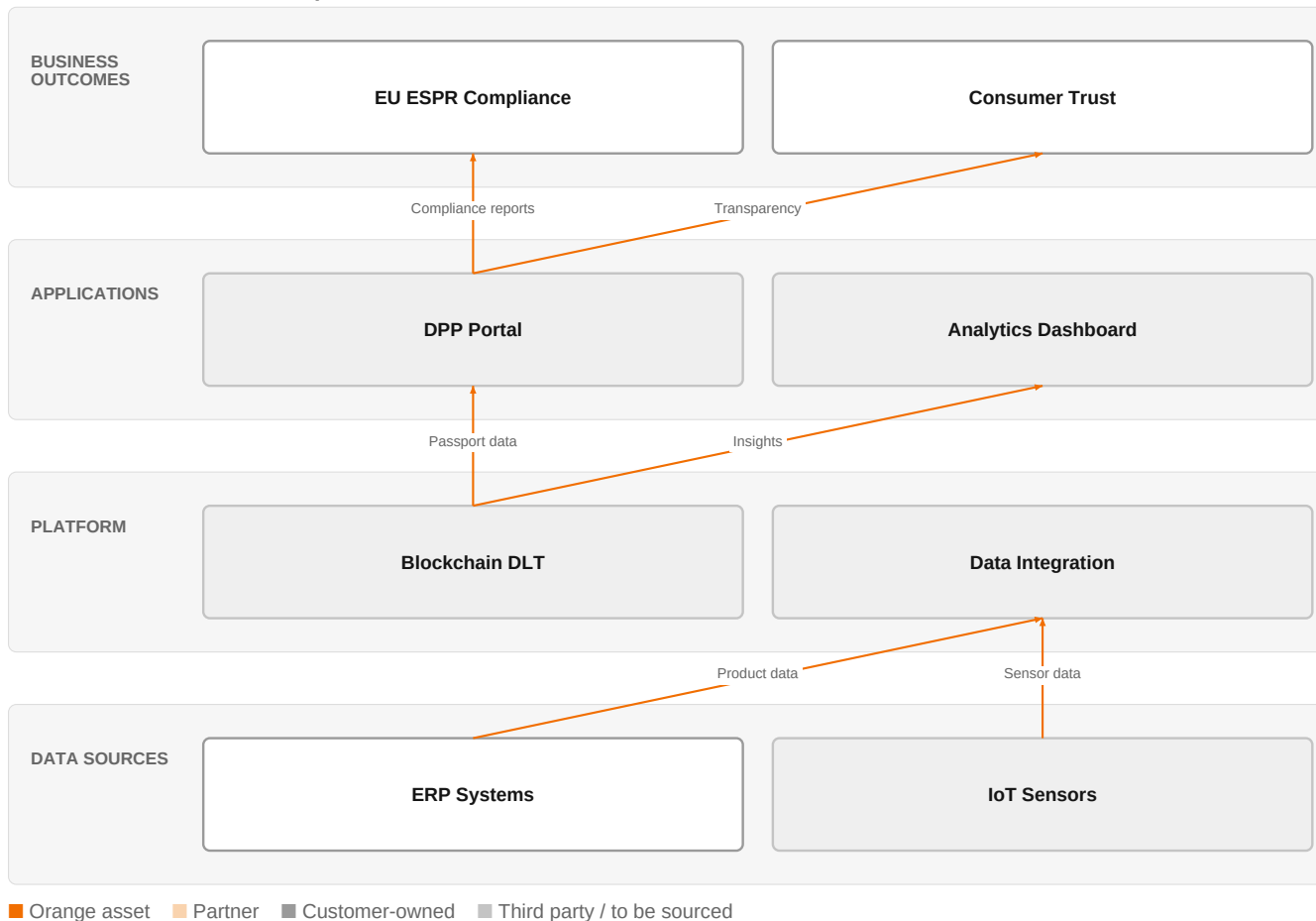
Bottom-up: enterprises x adoption x contract value

Factor	Value	Basis	Source	As of
Enterprises in Retail (EU27_2020)	195,962 enterprises	observed	Eurostat · sbs_sc_ovw	2024
Enterprises adopting Blockchain and DLT	44.3 % of enterprises (10+ employees)	proxy	Eurostat · isoc_cicce_usen2 · E_CC	2025
Annual value of one engagement	€200k per enterprise per year	assumption	config/sizing.yaml · fallback_bands_eur	size-2026-08-a
Size-weighted buyer base	20,550 enterprise-equivalents at large-enterprise engagement value	assumption	config/sizing.yaml · size_class_value_weights	size-2026-08-a
Assumed obtainable share of the serviceable market	0.1 % of SAM	assumption	config/sizing.yaml · obtainable_share	size-2026-08-a

Read this before quoting the number. Adoption rate is a declared proxy. Cloud computing services by NACE Rev. 2 activity series E_CC is used as a proxy for Blockchain and DLT: No DLT series at NACE level. Contract value is a configured assumption, not observed. Only 0 matching tender notices, below the 6 needed for a robust median, so the configured band for this business domain is used. It is an assumption owned by innovation-radar-curator, not evidence. Engagement value is scaled by enterprise size class (10-19: x0.04, 20-49: x0.09, 50-249: x0.35, GE250: x1.0), because the observed contract value comes from large-organisation tenders.

The solution, in one picture

Blockchain DPP for EU Compliance



This diagram shows how Orange's blockchain DLT platform integrates with customer systems to deliver compliant and trustworthy product passports.

Boxes marked as Orange assets are validated against the linked business graph; a box the model claimed for Orange without a matching asset is shown as third party.

What has changed

The EU's Ecodesign for Sustainable Products Regulation (ESPR) mandates Digital Product Passports for a wide range of product categories, including textiles, creating a regulatory imperative for retailers [SIG-AA85D6CD0C67]. The European technical standards for DPPs are now in place [SIG-AF45FABC720F], and the EU Commission has launched the Digital Product Passport Registry, covering textiles [SIG-047B44B6DF54, SIG-C4998B3557D1, SIG-311947DFF285, SIG-370D30A66C8C]. This regulatory push is accompanied by growing consumer interest in DPPs as a next-generation eco-label [SIG-FC25886E23D0] and evidence that blockchain-based DPPs can mitigate consumer stress in online luxury markets [SIG-AC6FFC29A702]. However, challenges remain in environmental data generation and DPP readiness for circular garments [SIG-48D7CEFBFA9E], and industry bodies are calling for flexible rules [SIG-E0C0A0C99916].

Sources: SIG-AA85D6CD0C67 openalex.org 2026-02-27; SIG-AF45FABC720F RECYCLING magazin 2026-08-03; SIG-047B44B6DF54 PubAffairs Bruxelles 2026-07-22; SIG-C4998B3557D1 INSIGHT EU MONITORING 2026-07-22; SIG-311947DFF285 Fibre2Fashion 2026-07-21; SIG-370D30A66C8C EU NEIGHBOURS east 2026-07-21

Who buys, and why now

The primary buyer is the Chief Sustainability Officer or Head of Compliance at a textile or luxury retailer selling into the EU. The pain is operational: they must generate and manage product lifecycle data to meet ESPR requirements, but their current systems rely on static label data that cannot track complex manufacturing processes or verify environmental performance [SIG-72777CF4A6C1]. The trigger is the imminent enforcement of ESPR and the live DPP Registry, which forces them to act now to avoid market access barriers.

Why it is hot now — every claim bound to a dated source

- DPPs in fashion and cosmetics support circular and collaborative ecosystems by enhancing transparency and traceability. [SIG-6B43CC88BC8E]

- DPPs enhance transparency, promote sustainable consumption, and support circular economy practices in the fashion industry. [SIG-FE1FBF23480C]
- DPPs can become the next-generation eco-label, analyzing consumer preferences for labeling. [SIG-FC25886E23D0]
- The EU's forthcoming DPP will create a dynamic, data-rich digital record for textile products. [SIG-72777CF4A6C1]
- Blockchain-based DPPs are being used in the luxury market to mitigate consumer stress. [SIG-AC6FFC29A702]
- The EU's forthcoming Digital Product Passport is set to create a dynamic, data-rich digital record for textile products, addressing supply chain transparency challenges. [SIG-72777CF4A6C1]
- Blockchain-based Digital Product Passports are used to mitigate consumer stress in online luxury markets, indicating a consumer-facing role. [SIG-AC6FFC29A702]
- Blockchain-based DPPs are being explored to mitigate consumer stress in online luxury markets, indicating retail relevance. [SIG-AC6FFC29A702]

What Orange would deliver, and why it can win

What Orange would deliver

Orange would assemble a solution using its Blockchain and DLT technology, supported by its AI/Data/Cloud experts, IoT experts, and Digital experts. The engagement would start with a discovery phase to map the customer's supply chain and data sources, then design a DPP schema aligned with EU standards. Orange would deploy a blockchain-based DPP platform that integrates with the customer's existing ERP and IoT sensors to capture product data, and provide a consumer-facing interface for DPP access. The solution would be piloted on a product line, then scaled across the portfolio.

Why Orange can win

Orange can win because it combines its own Blockchain and DLT technology with deep expertise in AI, data, cloud, and IoT, all of which are necessary to build a robust DPP solution. Orange also has experience in retail through its reference at Boulanger, demonstrating its ability to deliver in this sector. The trust and sovereignty angle is a differentiator, as Orange can offer sovereign, certified infrastructure, which is critical for handling sensitive product data.

Named assets behind this — each individually inspectable (LK-08)

Asset	Type	Link	Why it is linked	Curator
Blockchain and DLT	technology	L3	no offer path, but adjacent assets or Orange research exist	unconfirmed
AI / Data / Cloud experts	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
IoT experts	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
Digital experts	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
Orange Group researchers	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
Boulanger	reference	SUP	published reference in this vertical, different use case	unconfirmed

Competitive landscape

LOW competitive intensity (score 2.0; 0 of 8 listed players appear in this topic's own evidence)

One or two adjacent players. Orange would be shaping the space rather than fighting for it.

Competitor	Category	Position here	Basis	Evidence
Accenture	Global systems integrator	active in Retail; competes in CX: Customer Experience, OX: Smart Industries	structural	—
Telefonica Tech	Telecom operator peer	active in Retail	structural	—
Vodafone Business	Telecom operator peer	active in Retail; competes in OX: Smart Industries	structural	—
AWS	Hyperscaler	competes in OX: Smart Industries — also an Orange partner	structural	—

Google Cloud	Hyperscaler	competes in CX: Customer Experience — also an Orange partner	structural	—
Microsoft	Hyperscaler	competes in CX: Customer Experience — also an Orange partner	structural	—
Capgemini	Global systems integrator	competes in CX: Customer Experience, OX: Smart Industries	structural	—
Deutsche Telekom / T-Systems	Telecom operator peer	competes in OX: Smart Industries	structural	—

evidenced means this topic's own sources name the competitor — the signal ids are in the last column and are dated and clickable in the radar. **structural** means the curated register says they sell this technology into this vertical, which is true but is not proof they are in this deal. Register version competitors-2026-08-a.

Qualifying questions for the first meeting

- 1 What is your current approach to complying with the EU's ESPR for textiles, and what gaps have you identified in your product data management?
- 2 How do you currently capture and share product lifecycle data across your supply chain, and what systems are in place?
- 3 Have you started any pilot projects for Digital Product Passports, and if so, what were the key learnings?
- 4 Who in your organization is responsible for sustainability compliance, and how are they involved in technology procurement decisions?
- 5 What is your timeline for achieving ESPR compliance, and what budget has been allocated for this initiative?

Objections you will meet

They will say	You can answer
We already have a traceability system in place; why do we need a blockchain-based DPP?	That's a fair point. Many existing systems rely on static data that can't capture the full lifecycle or verify sustainability claims. A blockchain-based DPP provides a tamper-evident, shared record that meets the EU's requirements for dynamic, data-rich product information and can enhance consumer trust.
The regulatory requirements are still unclear; we prefer to wait.	It's true that some details are still evolving, but the EU has already launched the DPP Registry and technical standards are in place. Starting now allows you to pilot and refine your approach before mandatory enforcement, avoiding a last-minute scramble.
We are concerned about the cost and complexity of integrating DPPs into our existing systems.	Cost and complexity are valid concerns. Our approach is to start with a pilot on a specific product line, using our expertise to integrate with your existing ERP and IoT infrastructure. This minimizes disruption and allows you to see value before scaling.

Risks and unknowns

A key risk is that the regulatory landscape is still evolving, with industry bodies calling for flexible rules [SIG-E0C0A0C99916], which could change the requirements. Another risk is the challenge of generating reliable environmental data across complex supply chains [SIG-48D7CEFBFA9E]. It is also unknown how quickly consumers will adopt DPPs and whether they will truly influence purchasing decisions. Finally, the interoperability of DPP systems across different platforms and standards is not yet fully resolved.

Next action, by role (FR-17)

Role	Do this next
Presales / Proposal	Prepare a bid brief for a textile or luxury retailer's DPP initiative, leveraging Orange's Blockchain and DLT (L3) technology and IoT experts (SUP), and include a case study from Boulanger (SUP) to demonstrate our track record in similar digital transformations.
Sales	In a conversation with a European textile or luxury retail brand's sustainability or digital transformation lead, mention Orange's expertise in blockchain and DLT (L3) and our ability to support EU ESPR compliance through digital product passports, referencing our work with Boulanger (SUP).
Strategist / Innovator	Deep-dive into how Orange's Blockchain and DLT (L3) capability can be paired with IoT experts (SUP) to build a DPP prototype for textile supply chains, focusing on data interoperability and consumer trust features.

Evidence

Every claim in this brief resolves to one of these. Tier 1 is an authoritative primary source; tier 4 is vendor-published material, discounted in scoring (§4.3.7).

Signal	Date	Publisher	T	Title and link
SIG-AC6FFC29A702	2026-06-01	openalex.org	1	Positive Technology for Consumer WellBeing: The Role of BlockchainBased Digital Product Passp...
SIG-72777CF4A6C1	2026-04-28	Textile & Leather Review	1	A Sustainable Product Tracking System under the Combination of Textile Label Regulations and Di...
SIG-6B43CC88BC8E	2026-03-14	openalex.org	1	Integrating knowledge management and digital product passports to foster sustainable and collab...
SIG-85F36B96E175	2026-03-01	openalex.org	1	The Integration of RFID and Blockchain Technology for the Supply Chain Traceability of Durians
SIG-AA85D6CD0C67	2026-02-27	openalex.org	1	Digital product passports
SIG-1C2398920027	2026-02-04	openalex.org	1	Enhancing fruit supply chain traceability through blockchain and cryptographic protocols for ac...
SIG-BA196FF05244	2026-02-03	Springer Science and Busi...	1	Digital Product Passport for Circular Economy: A Review of Semantic Web Vocabularies and Requir...
SIG-FE1FBF23480C	2026-02-01	openalex.org	1	Digital product passports for cleaner production: Economic evidence from producers and consumer...
SIG-FC25886E23D0	2025-12-29	openalex.org	1	Can Digital Product Passports become the next-generation eco-label? Analyzing consumer preferen...
SIG-06EF21FC3A70	2025-12-04	openalex.org	1	Consumer Acceptance of Digital Product Passports: The Roles of Technological Awareness and Valu...
SIG-89C4D42B6B27	2025-04-23	MDPI AG	1	Digital Product Passport—EU Sustainability and Circularity Regulation
SIG-AF45FABC720F	2026-08-03	RECYCLING magazin	2	Digital Product Passport: European Technical Standards Now in Place - RECYCLING magazin
SIG-047B44B6DF54	2026-07-22	PubAffairs Bruxelles	2	Digital Product Passport Registry goes live - PubAffairs Bruxelles
SIG-C4998B3557D1	2026-07-22	INSIGHT EU MONITORING	2	EU Commission launches Digital Product Passport Registry - INSIGHT EU MONITORING
SIG-311947DFF285	2026-07-21	Fibre2Fashion	2	EU's digital product passport registry goes live, covering textiles - Fibre2Fashion
SIG-370D30A66C8C	2026-07-21	EU NEIGHBOURS east	2	EU's Digital Product Passport Registry goes live - EU NEIGHBOURS east
SIG-48D7CEFBFA9E	2026-07-20	Stockholm Environment Ins...	2	From policy to practice: challenges of environmental data generation and Digital Product Passpo...
SIG-E0C0A0C99916	2026-06-22	Textile Today	2	Ecommerce Europe calls for flexible Digital Product Passport rules for apparel sector - Textile...
SIG-4FE0689B82DF	2026-04-30	Packaging Digest	2	EU Digital Product Passport Impact on US Supply Chains - Packaging Digest

How this brief was made

Computed, not written. Attractiveness, right to win, competitive intensity and every figure in the market-opportunity section are computed from stored inputs and can be re-derived (DR-05, NFR-01). No language model produced a number anywhere in this document (§4.4.4).

Written, then checked. The narrative sections and the solution diagram were generated from this topic's evidence and from the named asset and competitor lists. Factual sections must cite signal ids attached to this topic; sections that failed that check, or that contained a generated quantity or an unsupplied organisation, were removed rather than rewritten.

Removed by those checks in this brief: competitive_landscape (no claim survived evidence binding); diagram (4 box(es) claimed to be an Orange asset that was not supplied — shown as third-party instead)

What	Version	When
Scores — weight set	w-2026-08-a	2026-08-18T19:08:26+00:00
Market sizing — assumptions	size-2026-08-a	2026-08-18T21:06:15+00:00
Competitor register	competitors-2026-08-a	2026-08-18T21:06:18+00:00
Narrative — prompt / model	describe-v1 / deepseek-chat	2026-08-18T20:42:54+00:00
Pipeline	0.1.0	2026-08-18

Generated 2026-08-19 04:58 UTC. Scores computed under a different weight set are not comparable with these (SC-10), and neither are market sizes computed under a different sizing version.